DEPi | Digital Egypt Pioneers Initiative

Track: AI Healthcare Professional

# Smart Clinical Quality & Incident Surveillance System

AI-driven safety. Faster decisions. Better patient outcomes.

**Project Overview**
- AI Healthcare
- Quality & Safety
- OVR + Medication

→ **Business Problem**
- Medication errors
- Slow OVR processing
- Incomplete reports

→ **Solution**
- AI automation
- PHI Gate
- Human-in-the-loop

→ **Core Features**
- PHI filtering
- RCA & severity
- CAPA + alerts

→ **Architecture**
- 2 Workflows
- LLM + tools
- LangSmith tracing

→ **AI Components**
- LLM
- RCA & classification
- Prompt engineering

→ **Tech Stack**
- Python, n8n
- LangSmith
- Docker, PostgreSQL

→ **Timeline**
- Plan → Design
- Build → Test
- Deploy

→ **Deployment**
- Cloud
- Scalable
- Secure

→ **Deliverables**
- Working system
- Dashboard
- Documentation

→ **Future Enhancements**
- Voice-to-text
- Predictive dashboard
- More integrations

## Business Impact

**Patients**
Error-free medication plan

**Clinical Staff**
Automated drug-safety screening

**Quality Managers**
Real-time OVR triage & insights

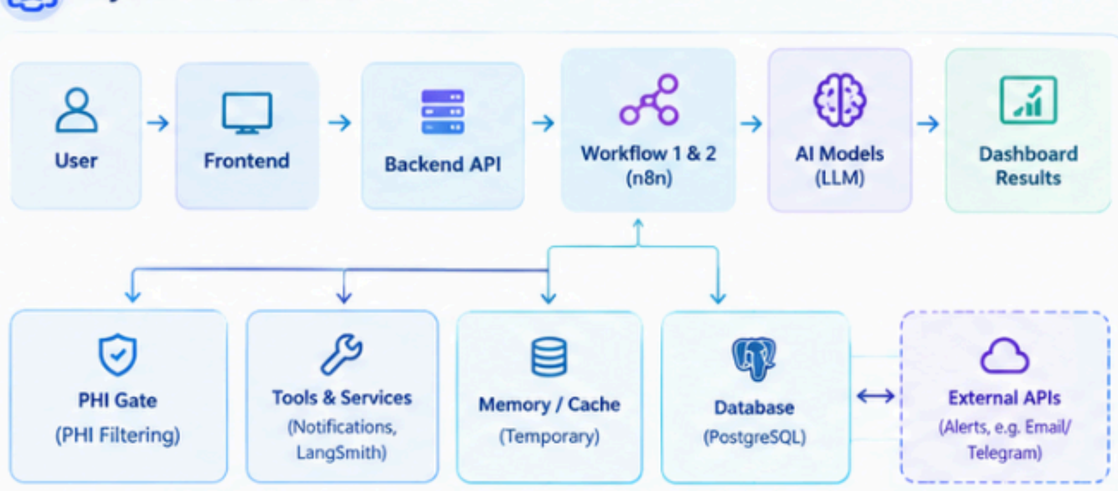
## Target Audience

**Users**
Pharmacists
Nurses
Quality Officers

**Beneficiaries**
Hospitalized patients
Post-discharge individuals

**Buyers**
CMO
Quality Director

## System Architecture




## AI Components




## Technology Stack

Python · FastAPI · n8n · LangSmith · Docker · PostgreSQL · OpenAI · GitHub

## Development Timeline

Planning → Design → Implementation → Testing → Deployment

## Project Outcomes

- Working System
- Dashboard
- Source Code
- Documentation
- REST API
- AI Model
- Deployment
- Web Application